

Alternative Grading in Mathematics Courses

Practices, Outcomes, and Challenges

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Wabash.

Outline

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Introduction

Origin Story

- Tension
 - Opportunities to learn
 - Ensuring understanding
- Allowed students to recover points from exams
- Changing relative weighting
- Geometric mean
- Challenge:
 - Recover from early mistakes
 - Grades reflect real understanding

Initial Approach

Initial Approach

- Specifications Grading
- Projects due each week
- Four levels for projects:
 - E: Excellent
 - S: Satisfactory
 - P: Partial understanding
 - I: Incomplete
- Oops tokens

CSC 111 Fall 2018

CSC 111 Fall 2018 Grading Scheme

	A	B	C	D	F
Class meetings (28 total)	26	24	22	20	-
Classroom activities (28)	26	24	22	20	-
Chapter reading quizzes	90%	80%	70%	60%	-
Projects: S or E (14 total)	12	11	10	9	-
Projects: E	9	7	5	-	-

MAT 112 Spring 2019

MAT 112 Spring 2019 Grading Scheme

	A	B	C	D	F
Basic Skills Exam				Pass	-
Exam average	90%	80%	70%	60%	-
Projects: S or E			75%	60%	-
Projects: E	50%	25%	-	-	-
Homework: S or E			75%	60%	-
Homework: E	50%	25%	-	-	-

CSC 111 Fall 2020

Simplified Approach

- Two levels for projects: Pass or Not
- Submit any two projects each week

CSC 111 Fall 2020 Grading Scheme

	A	B	C	D	F
Projects (14 total)	13	12	10	9	-

MAT 225 Fall 2020 – Standards-Based Testing

Standards-Based Testing

- Classes focused on learning targets
- Students demonstrate mastery on exams (i.e., “mastery opportunities”)
- Students can work on any learning target not yet mastered

MAT 225 Fall 2020 Grading Scheme

	A	B	C	D	F
Learning Targets (32 total)	29	26	23	20	-
Online Homework Points (1181 total)			800	700	-

Mixed Approach

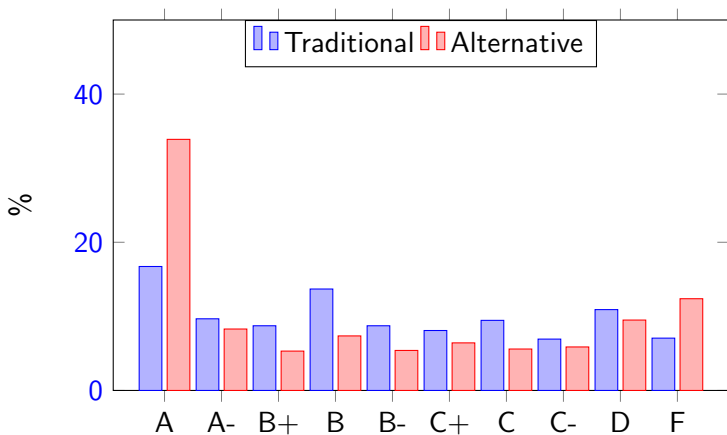
MAT 112 Fall 2025 Grading Scheme

	A	A-	B+	B	B-	C+	C	C-	D
Learning Targets (36)	34	33	32	30	29	28	27	26	22
Projects (7)	6			5			4		3
Online Homework (400)							300		240
Mathematical Life							1		
Resource Utilization									1

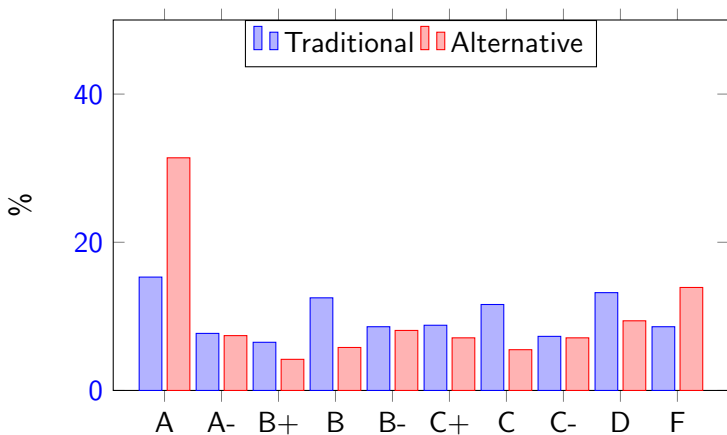
MAT 331 Spring 2026 Grading Scheme

	A	A-	B+	B	B-	C+	C	C-	D
Learning Targets (37)	34	33	32	31	30	28	27	26	22
Writing Assignments (13)	12		11		10		9	8	7
Mathematical Life	6		5		4		3	2	1

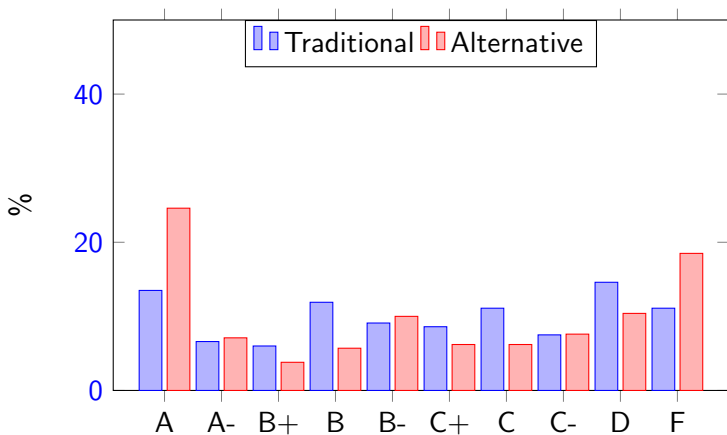
All Courses Grade Distribution



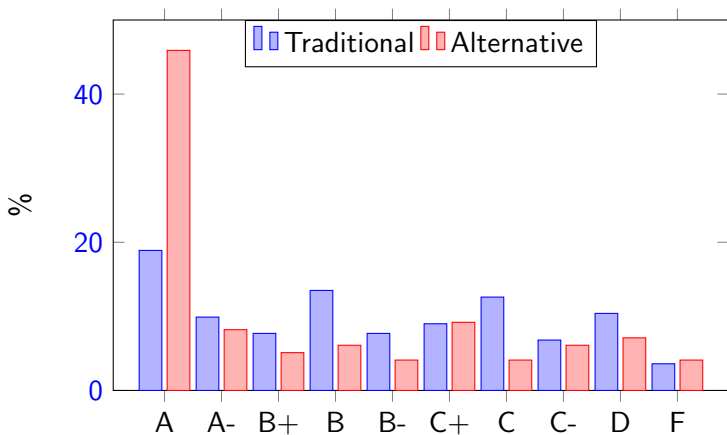
Mathematics Grade Distribution



Lower Level Courses Grade Distribution



Upper Level Grade Distribution



AI

AI resistant?

- Grades not averaged together
- Performance on projects cannot balance out low exam scores
- Must still perform well on in-class assessments

Disadvantages

Disadvantages

- Procrastination
- Declining attendance in recent years

Possible Modifications

Possible Modifications

- Required final exam
- Attendance policy
- In-class assignments

Summary

Summary

My experience suggests that alternative grading can provide a more accurate picture of student learning in mathematics courses, particularly when course grades depend on demonstrated mastery rather than accumulated points. At the same time, these systems require careful design to address student procrastination, attendance, and logistical complexity. Continued refinement of these approaches may help preserve their benefits while reducing their drawbacks.